

## 2. Randomization Inference

ISS5096 || ECI

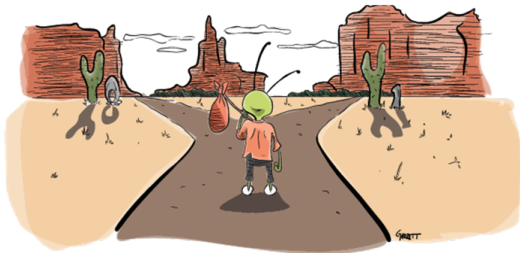
Jaewon (“Jay-one”) Yoo

National Tsing Hua University

# Outline

1. Randomized Experiments
2. Randomization Inference
3. Application of Randomization Inference
4. In-Class Exercise

# Where are we? Where are we going?



Source: Chapter 1 of *Mastering Metrics (Textbook 2)* by J. Angrist & J. Pischke

- Last time: defining causal effects as **contrasts of counterfactuals**.
- What can we learn about these contrasts from randomized experiments?
  - Message: randomization allows for inference under practically no assumptions.
- Useful to have notations for vector of all r.v.s:
  - Treatment:  $\mathbf{D} = \{D_1, D_2, \dots, D_n\}$
  - Potential outcomes:  $\mathbf{Y}(1) = \{Y_1(1), \dots, Y_n(1)\}$
  - Covariates:  $\mathbf{X} = \{X_1, \dots, X_n\}$

# 1/ Randomized Experiments

# Randomized Experiments

- **Experiment:** method/design where the researcher controls the treatment assignment.
  - $p_i$  (i.e., the probability of treatment assignment) is controlled by and known to the researcher in an experiment.

# Randomized Experiments

- **Experiment:** method/design where the researcher controls the treatment assignment.
  - $p_i$  (i.e., the probability of treatment assignment) is controlled by and known to the researcher in an experiment.
- **Randomized experiment** is an experiment with two properties:
  1. **Positivity:** assignment is probabilistic (i.e.,  $0 < p_i < 1$ ).
    - No deterministic assignment.
  2. **Unconfoundedness:**  $\mathbb{P}[D_i = 1 | Y(1), Y(0)] = \mathbb{P}[D_i = 1]$ 
    - Treatment assignment does not depend on any potential outcomes.
    - If patients were assigned to treatment group based on how researchers anticipate the patients will respond to the medication?

# Context: Effect of AI Assistant on Learning Outcomes

- Q: Does AI assistant help students achieving higher academic performance?
  - Difficult with observational studies: usage of AI assistant correlated with lots of stuff!
- Randomized controlled trial can be helpful.
- Setup:
  - Units: students  $i$
  - Treatment: assignment to a tutor with AI assistant ( $D_i = 1$ ) or not ( $D_i = 0$ )
  - Outcome: student passes a standardized test ( $Y_i = 1$ ) or not ( $Y_i = 0$ )
- If AI assistant  $\rightsquigarrow$  performance, we should see a difference between the treatment and control groups.

# Why Randomize?

- Randomization makes treated and control groups **comparable!**
  - If both groups are random samples from all units in the study.
  - $\rightsquigarrow$  **Balanced** on all variables: roughly,  $N_{men}^T \approx N_{men}^C$ , etc.
  - True for all observed & unobserved pretreatment variables.
  - Most importantly: potential outcomes are comparable by unconfoundedness:

$$\mathbb{P}(Y_i(1) = 1 | D_i = 1) = \mathbb{P}(Y_i(1) = 1) = \mathbb{P}(Y_i(1) = 1 | D_i = 0)$$



# Why Randomize?

- Randomization makes treated and control groups **comparable!**
  - If both groups are random samples from all units in the study.
  - $\rightsquigarrow$  **Balanced** on all variables: roughly,  $N_{men}^T \approx N_{men}^C$ , etc.
  - True for all observed & **unobserved pretreatment** variables.
  - Most importantly: potential outcomes are comparable by unconfoundedness:

$$\mathbb{P}(Y_i(1) = 1 | D_i = 1) = \mathbb{P}(Y_i(1) = 1) = \mathbb{P}(Y_i(1) = 1 | D_i = 0)$$

- Caveat: groups are not comparable on **post-treatment** variables.
  - Recall consistency:  $Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$
  - $Y_i(1) \perp\!\!\!\perp D_i$  but not  $Y_i \perp\!\!\!\perp D_i$

# Why Randomize?

- Randomization makes treated and control groups **comparable!**
  - If both groups are random samples from all units in the study.
  - $\rightsquigarrow$  **Balanced** on all variables: roughly,  $N_{men}^T \approx N_{men}^C$ , etc.
  - True for all observed & **unobserved pretreatment** variables.
  - Most importantly: potential outcomes are comparable by unconfoundedness:

$$\mathbb{P}(Y_i(1) = 1 | D_i = 1) = \mathbb{P}(Y_i(1) = 1) = \mathbb{P}(Y_i(1) = 1 | D_i = 0)$$

- Caveat: groups are not comparable on **post-treatment** variables.
  - Recall consistency:  $Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$
  - $Y_i(1) \perp\!\!\!\perp D_i$  but not  $Y_i \perp\!\!\!\perp D_i$
- Really talking about **ideal** randomized experiment:
  - Full compliance, no missing data.
  - Important to admit limitations: *external validity (i.e., generalizability), sample selection, Hawthorne effect*, etc.

# Types of Experiments

- Experiments can be classified by their **assignment mechanism**.
  - What (random) function do we use to assign treatment?

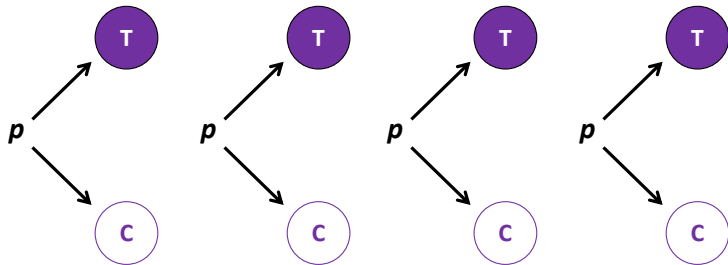
# Types of Experiments

- Experiments can be classified by their **assignment mechanism**.
  - What (random) function do we use to assign treatment?
- **Bernoulli (coin flips) experiment:**
  - Each unit is assigned  $D_i = 1$  with prob.  $p_i$  independently.
  - Downside: “bad” randomization is possible (i.e., all assigned into treatment or control group)

# Types of Experiments

- Experiments can be classified by their **assignment mechanism**.
  - What (random) function do we use to assign treatment?
- **Bernoulli (coin flips) experiment:**
  - Each unit is assigned  $D_i = 1$  with prob.  $p_i$  independently.
  - Downside: “bad” randomization is possible (i.e., all assigned into treatment or control group)
- **Completely randomized experiment:**
  - Randomly sample  $n_1$  units from the population to be treated.
  - For any given  $i$ ,  $p_i = \mathbb{P}(D_i = 1) = \frac{n_1}{n}$

# Bernoulli Assignment



# Completely Randomized Design



- Begin with  $N = 6$ , and say, we want to have  $N_t = 3$
- Randomly pick 3 from  $\{1, 2, 3, 4, 5, 6\}$ : 2, 4, 5
- Fixed number of treated units induces dependence between  $D_i$  and  $D_j$ 
  - Knowing unit 2 is treated  $\rightsquigarrow$  unit 3 is less likely to be treated.
  - Makes variance calculations tricky (we will come back to this)

# Completely Randomized Design



- Begin with  $N = 6$ , and say, we want to have  $N_t = 3$
- Randomly pick 3 from  $\{1, 2, 3, 4, 5, 6\}$ : 2, 4, 5
- Fixed number of treated units induces dependence between  $D_i$  and  $D_j$ 
  - Knowing unit 2 is treated  $\rightsquigarrow$  unit 3 is less likely to be treated.
  - Makes variance calculations tricky (we will come back to this)
- We can also randomize within groups (**block/stratified randomization**).
  - When we have 2 units in each block, this is referred to as a **pair-matched design**.



# Example Data from AI RCT

| Student | AI Assistant | Pass the Exam? | $Y_i(0)$ | $Y_i(1)$ |
|---------|--------------|----------------|----------|----------|
|         | $D_i$        | $Y_i$          |          |          |
| 1       | 1            | 0              | ?        | 0        |
| 2       | 1            | 0              | ?        | 0        |
| 3       | 0            | 1              | 1        | ?        |
| 4       | 1            | 0              | ?        | 0        |
| 5       | 1            | 1              | ?        | 1        |
| 6       | 0            | 1              | 1        | ?        |
| 7       | 0            | 0              | 0        | ?        |
| 8       | 1            | 1              | ?        | 1        |
| 9       | 0            | 1              | 1        | ?        |
| 10      | 0            | 0              | 0        | ?        |

- Students passed the exam 2/5 times with AI assistant vs. 3/5 times w/o AI assistant!
- Very small sample size  $\rightsquigarrow$  can we learn anything from this data?

## **2/** Randomization Inference

# What is Randomization Inference?

- **Randomization inference (RI):** Inference based on different possible randomization of treatment.

# What is Randomization Inference?

- **Randomization inference (RI):** Inference based on different possible randomization of treatment.
  - R. Fisher: randomization is the “reasoned basis for inference.”
  - We can generate exact p-values for tests of a “sharp” null hypothesis.
  - Also called: **design-based inference**.

# What is Randomization Inference?

- **Randomization inference (RI):** Inference based on different possible randomization of treatment.
  - R. Fisher: randomization is the “reasoned basis for inference.”
  - We can generate exact p-values for tests of a “sharp” null hypothesis.
  - Also called: **design-based inference**.
- Allows us to make **exact, distribution-free** inferences.
  - No reliance on normality, etc.
  - No reliance on large-sample approximations.
  - $\rightsquigarrow$  non-parametric, but less flexible.

# Brief Review of Hypothesis Testing

- RI is a type of hypothesis testing in a randomized experiment, so it is helpful to review!

# Brief Review of Hypothesis Testing

- RI is a type of hypothesis testing in a randomized experiment, so it is helpful to review!

1. *Choose a null hypothesis:*

- $H_0 : \beta_1 = 0$
- Claim  $\Rightarrow$  no average treatment effect.
- This is a claim we would like to reject.

# Brief Review of Hypothesis Testing

- RI is a type of hypothesis testing in a randomized experiment, so it is helpful to review!

1. *Choose a null hypothesis:*

- $H_0 : \beta_1 = 0$
- Claim  $\Rightarrow$  no average treatment effect.
- This is a claim we would like to reject.

2. *Choose a test statistic:*

- e.g.,  $Z_i = (X_i - \bar{X}) / (s / \sqrt{n})$



# Brief Review of Hypothesis Testing

- RI is a type of hypothesis testing in a randomized experiment, so it is helpful to review!

1. *Choose a null hypothesis:*

- $H_0 : \beta_1 = 0$
- Claim  $\Rightarrow$  no average treatment effect.
- This is a claim we would like to reject.

2. *Choose a test statistic:*

- e.g.,  $Z_j = (X_j - \bar{X}) / (s / \sqrt{n})$

3. *Determine the distribution of the test statistic under the null.*

- Statistical thought experiment: if we knew the truth, what data should we expect?

# Brief Review of Hypothesis Testing

- RI is a type of hypothesis testing in a randomized experiment, so it is helpful to review!

1. *Choose a null hypothesis:*

- $H_0 : \beta_1 = 0$
- Claim  $\Rightarrow$  no average treatment effect.
- This is a claim we would like to reject.

2. *Choose a test statistic:*

- e.g.,  $Z_j = (X_j - \bar{X}) / (s / \sqrt{n})$

3. *Determine the distribution of the test statistic under the null.*

- Statistical thought experiment: if we knew the truth, what data should we expect?

4. *Calculate the probability of the test statistics under the null.*

- What is this called? **p-value**

# Sharp Null Hypothesis of No Effect

- **Sharp null hypothesis:**

$$H_0 : \tau_i = Y_i(1) - Y_i(0) = 0 \quad \text{for all } i \quad (1)$$

- Specified for each unit  $i$  & assumes zero effect for every unit (hence, *sharp* vs. *weak* hypothesis, which assumes ATE = 0.)

# Sharp Null Hypothesis of No Effect

- **Sharp null hypothesis:**

$$H_0 : \tau_i = Y_i(1) - Y_i(0) = 0 \quad \text{for all } i \quad (1)$$

- Specified for each unit  $i$  & assumes zero effect for every unit (hence, *sharp* vs. *weak* hypothesis, which assumes ATE = 0.)
- What if treatment affected no one at all?

# Sharp Null Hypothesis of No Effect

- **Sharp null hypothesis:**

$$H_0 : \tau_i = Y_i(1) - Y_i(0) = 0 \quad \text{for all } i \quad (1)$$

- Specified for each unit  $i$  & assumes zero effect for every unit (hence, *sharp* vs. *weak* hypothesis, which assumes  $ATE = 0$ .)
- What if treatment affected no one at all?
- Implies no **average** treatment effect, but no  $ATE \Rightarrow$  sharp null.
  - Take a simple example with only two units:  $\tau_1 = 1$  and  $\tau_2 = -1$ .
  - Here,  $\tau = 0$ , but the sharp null is violated.

# Sharp Null Hypothesis of No Effect

- **Sharp null hypothesis:**

$$H_0 : \tau_i = Y_i(1) - Y_i(0) = 0 \quad \text{for all } i \quad (1)$$

- Specified for each unit  $i$  & assumes zero effect for every unit (hence, *sharp* vs. *weak* hypothesis, which assumes ATE = 0.)
- What if treatment affected no one at all?
- Implies no **average** treatment effect, but no ATE  $\Rightarrow$  sharp null.
  - Take a simple example with only two units:  $\tau_1 = 1$  and  $\tau_2 = -1$ .
  - Here,  $\tau = 0$ , but the sharp null is violated.
- If the sharp null is true, then we know all the potential outcomes:

$$Y_i(1) = Y_i(0) = Y_i \quad (2)$$

# Life Under the Sharp Null

We can use the sharp null (i.e.,  $Y_i(1) = Y_i(0) = Y_i$ ) to fill in the missing potential outcomes:

| Student | AI Assistant<br>$D_i$ | Pass the Exam?<br>$Y_i$ | $Y_i(0)$ | $Y_i(1)$ |
|---------|-----------------------|-------------------------|----------|----------|
| 1       | 1                     | 0                       | ?        | 0        |
| 2       | 1                     | 0                       | ?        | 0        |
| 3       | 0                     | 1                       | 1        | ?        |
| 4       | 1                     | 0                       | ?        | 0        |
| 5       | 1                     | 1                       | ?        | 1        |
| 6       | 0                     | 1                       | 1        | ?        |
| 7       | 0                     | 0                       | 0        | ?        |
| 8       | 1                     | 1                       | ?        | 1        |
| 9       | 0                     | 1                       | 1        | ?        |
| 10      | 0                     | 0                       | 0        | ?        |

# Life Under the Sharp Null

We can use the sharp null (i.e.,  $Y_i(1) = Y_i(0) = Y_i$ ) to fill in the missing potential outcomes:

| Student | AI Assistant<br>$D_i$ | Pass the Exam?<br>$Y_i$ | $Y_i(0)$ | $Y_i(1)$ |
|---------|-----------------------|-------------------------|----------|----------|
| 1       | 1                     | 0                       | 0        | 0        |
| 2       | 1                     | 0                       | 0        | 0        |
| 3       | 0                     | 1                       | 1        | 1        |
| 4       | 1                     | 0                       | 0        | 0        |
| 5       | 1                     | 1                       | 1        | 1        |
| 6       | 0                     | 1                       | 1        | 1        |
| 7       | 0                     | 0                       | 0        | 0        |
| 8       | 1                     | 1                       | 1        | 1        |
| 9       | 0                     | 1                       | 1        | 1        |
| 10      | 0                     | 0                       | 0        | 0        |



# Test Statistic

## Test statistic

A test statistic is a known, scalar quantity calculated from the treatment assignments, observed outcomes, and possibly covariates:  $T(\mathbf{D}, \mathbf{Y}, \mathbf{X})$ .

- Test statistics measure how unusual the data is under the null.
- Typically measures the relationship between two variables (in causal inference).
- We want a test statistic with high **statistical power**:
  - Has large values when the null is likely false.
  - These large values are unlikely when the null is true.
- These will help us perform a test of the sharp null.

# Null/Randomization Distribution

- What is the distribution of the test statistic under the sharp null?
  - If there was no effect (i.e., sharp null), what test statistics would we expect over different randomizations?

# Null/Randomization Distribution

- What is the distribution of the test statistic under the sharp null?
  - If there was no effect (i.e., sharp null), what test statistics would we expect over different randomizations?
- **Key insight of RI:** Sharp null  $\rightsquigarrow$  treatment assignment doesn't matter.
  - Shuffling treatment vector won't change the outcomes!
  - $Y_i(1) = Y_i(0) = Y_i$

# Null/Randomization Distribution

- What is the distribution of the test statistic under the sharp null?
  - If there was no effect (i.e., sharp null), what test statistics would we expect over different randomizations?
- **Key insight of RI:** Sharp null  $\rightsquigarrow$  treatment assignment doesn't matter.
  - Shuffling treatment vector won't change the outcomes!
  - $Y_i(1) = Y_i(0) = Y_i$
- **Randomization distribution:** distribution of  $T$  under the sharp null.

# Calculate P-value

- How often would we get a test statistic this big or bigger if the sharp null holds?
- **Exact p-values:**

$$Pr(T \geq T^{obs}) = \frac{1}{K} \sum_{k=1}^K \mathbb{I}(T(\mathbf{d}, \mathbf{Y}, \mathbf{X}) \geq T^{obs}) \quad (3)$$

- How often  $T$  under different randomization larger than the  $T^{obs}$  divided by total number of randomizations ( $K$ )?
- Can be compared to a chosen threshold,  $\alpha$ , to determine whether to reject the sharp null.

# Randomization Inference Step-by-Step

1. Choose a sharp null hypothesis and a test statistic.
2. Calculate observed test statistic:  $T^{obs} = T(\mathbf{D}, \mathbf{Y}, \mathbf{X})$ .
3. Randomly select different treatment assignment vector  $\tilde{\mathbf{D}}_1$ .
  - e.g., if  $\mathbf{D} = \{1, 1, 1, 0, 0, 0\}$ , then  $\tilde{\mathbf{D}}_1$  could be  $\{1, 1, 0, 1, 0, 0\}$ .
4. Calculate  $\tilde{T}_1 = T(\tilde{\mathbf{D}}_1, \mathbf{Y}, \mathbf{X})$ .
5. Repeat **steps 3** and **4** to get  $\tilde{T} = \{\tilde{T}_1, \dots, \tilde{T}_K\}$  (randomization dist.)
6. Calculate the p-value:  $p = \frac{1}{K} \sum_{k=1}^K \mathbb{I}(\tilde{T}_k \geq T^{obs})$ .

# Difference in Means as a Test Statistic

- Many different types of test statistics with different strengths/benefits.

# Difference in Means as a Test Statistic

- Many different types of test statistics with different strengths/benefits.
- Natural (if not optimal): absolute difference-in-means estimator:

$$T_{\text{diff}} = \left| \frac{1}{n_1} \sum_{i=1}^N D_i Y_i - \frac{1}{n_0} \sum_{i=1}^N (1 - D_i) Y_i \right| \quad (4)$$

- Larger values of  $T_{\text{diff}}$  are evidence against the sharp null.



# Difference in Means as a Test Statistic

- Many different types of test statistics with different strengths/benefits.
- Natural (if not optimal): absolute difference-in-means estimator:

$$T_{\text{diff}} = \left| \frac{1}{n_1} \sum_{i=1}^N D_i Y_i - \frac{1}{n_0} \sum_{i=1}^N (1 - D_i) Y_i \right| \quad (4)$$

- Larger values of  $T_{\text{diff}}$  are evidence against the sharp null.
- A good estimator for constant treatment effects (across all units) with relatively few outliers in the potential outcomes.

## **3/** Application of Randomization Inference

## Example: Encouraging Donation to NTHU

- Suppose we are targeting and encouraging 6 people to make a donation to National Tsing Hua University.
- As an encouragement, we send 3 of them an email with inspirational stories of learning from our alumni.
- Afterwards, we observe them giving between \$0 and \$5.
- Simple example to show the steps of RI in a concrete case.

# Randomization Distribution

| Unit    | Email<br>$D_i$ | Donation in $US\$$<br>$Y_i$ | $Y_i(0)$ | $Y_i(1)$ |
|---------|----------------|-----------------------------|----------|----------|
| Brian   | 1              | 3                           | (3)      | 3        |
| Desi    | 1              | 5                           | (5)      | 5        |
| Medjine | 1              | 0                           | (0)      | 0        |
| Natasha | 0              | 4                           | 4        | (4)      |
| Fifi    | 0              | 0                           | 0        | (0)      |
| Matthew | 0              | 1                           | 1        | (1)      |

$$T_{\text{diff}} = |8/3 - 5/3| = 1$$

with the observed treatment assignment,  $D = \{1, 1, 1, 0, 0, 0\}$ .

# Randomization Distribution

| Unit    | Email<br>$\tilde{D}_i$ | Donation in $US\$$<br>$Y_i$ | $Y_i(0)$ | $Y_i(1)$ |
|---------|------------------------|-----------------------------|----------|----------|
| Brian   | 1                      | 3                           | (3)      | 3        |
| Desi    | 1                      | 5                           | (5)      | 5        |
| Medjine | 0                      | 0                           | 0        | (0)      |
| Natasha | 1                      | 4                           | (4)      | 4        |
| Fifi    | 0                      | 0                           | 0        | (0)      |
| Matthew | 0                      | 1                           | 1        | (1)      |

$$\tilde{T}_{\text{diff}} = |12/3 - 1/3| = 3.67 \text{ if } \tilde{D} = \{1, 1, 0, 1, 0, 0\}$$

$$\tilde{T}_{\text{diff}} = |8/3 - 5/3| = 1 \text{ if } \tilde{D} = \{1, 1, 1, 0, 0, 0\}$$

$$\tilde{T}_{\text{diff}} = |9/3 - 4/3| = 1.67 \text{ if } \tilde{D} = \{0, 1, 1, 1, 0, 0\}$$

# Randomization Distribution

| $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ | $D_6$ | Diff. in Means |
|-------|-------|-------|-------|-------|-------|----------------|
| 1     | 1     | 1     | 0     | 0     | 0     | 1.00           |
| 1     | 1     | 0     | 1     | 0     | 0     | 3.67           |
| 1     | 1     | 0     | 0     | 1     | 0     | 1.00           |
| 1     | 1     | 0     | 0     | 0     | 1     | 1.67           |
| 1     | 0     | 1     | 1     | 0     | 0     | 0.33           |
| 1     | 0     | 1     | 0     | 1     | 0     | 2.33           |
| 1     | 0     | 1     | 0     | 0     | 1     | 1.67           |
| 1     | 0     | 0     | 1     | 1     | 0     | 0.33           |
| 1     | 0     | 0     | 1     | 0     | 1     | 1.00           |
| 1     | 0     | 0     | 0     | 1     | 1     | 1.67           |
| 0     | 1     | 1     | 1     | 0     | 0     | 1.67           |
| 0     | 1     | 1     | 0     | 1     | 0     | 1.00           |
| 0     | 1     | 1     | 0     | 0     | 1     | 0.33           |
| 0     | 1     | 0     | 1     | 1     | 0     | 1.67           |
| 0     | 1     | 0     | 1     | 0     | 1     | 2.33           |
| 0     | 1     | 0     | 0     | 1     | 1     | 0.33           |
| 0     | 0     | 1     | 1     | 1     | 0     | 1.67           |
| 0     | 0     | 1     | 1     | 0     | 1     | 1.00           |
| 0     | 0     | 1     | 0     | 1     | 1     | 3.67           |
| 0     | 0     | 0     | 1     | 1     | 1     | 3.67           |

# Application in R

```
1 > library(ri) # loading randomization inference package, ri.
2 > y <- c(3, 5, 0, 4, 0, 1)
3 > D <- c(1, 1, 1, 0, 0, 0)
4
5 # Diff. in means as a test stat.
6 > T_obs <- abs(mean(y[D == 1]) - mean(y[D == 0]))
7
8 # genperms() to generate all possible treatment assignments, tilde D.
9 > D_bold <- ri::genperms(D) # 20 diff. ways
10 > D_bold[, 1:10]
```

| ## | [,1] | [,2] | [,3] | [,4] | [,5] | [,6] | [,7] | [,8] | [,9] | [,10] |
|----|------|------|------|------|------|------|------|------|------|-------|
| ## | 1    | 1    | 1    | 1    | 1    | 1    | 1    | 1    | 1    | 1     |
| ## | 1    | 1    | 1    | 1    | 0    | 0    | 0    | 0    | 0    | 0     |
| ## | 1    | 0    | 0    | 0    | 1    | 1    | 1    | 0    | 0    | 0     |
| ## | 0    | 1    | 0    | 0    | 1    | 0    | 0    | 1    | 1    | 0     |
| ## | 0    | 0    | 1    | 0    | 0    | 1    | 0    | 1    | 0    | 1     |
| ## | 0    | 0    | 0    | 1    | 0    | 0    | 1    | 0    | 1    | 1     |

# Application in R

```
1 # Calculating the test statistics to obtain the rand. dist.
2 > rdist <- rep(NA, times = ncol(D_bold))
3 > for (i in seq_len(ncol(D_bold))) {
4     D_tilde <- D_bold[,i]
5     rdist[i] <- abs(mean(y[D_tilde == 1]) - mean(y[D_tilde == 0]))
6 }
7 > round(rdist, digits = 3) # print randomization distribution
```

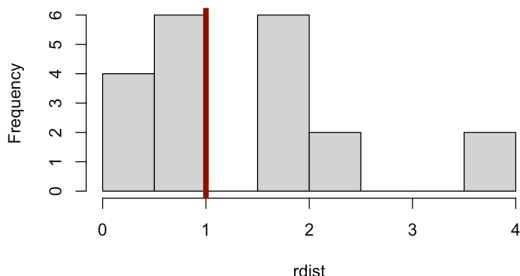
---

```
## [1] 1.000 3.667 1.000 1.667 0.333 2.333 1.667 0.333 1.000 1.667
## [11] 1.667 1.000 0.333 1.667 2.333 0.333 1.667 1.000 3.667 1.000
```



# Example: Computation in R Cont.

```
1 > hist(x = rdist) # visualize the randomization distribution
2 > abline(v = 1, col = "darkred", lwd = 5) # Red line = observed T
```



```
1 > mean(rdist >= T_obs) # Computing p-value
```

```
## [1] 0.8
```

- $\alpha$  levels? Not enough evidence to reject the sharp null of no effect! (or not enough evidence to say encouragement helped)

# Computational Burden in RI

Computing the exact randomization distribution is not always possible:

- $n = 6$  and  $n_1 = 3 \rightsquigarrow 20$  assignment vectors ( ${}^6C_3$ ).
- $n = 10$  and  $n_1 = 5 \rightsquigarrow 252$  vectors ( ${}^{10}C_5$ ).
- $n = 100$  and  $n_1 = 50 \rightsquigarrow 1.009 \times 10^{29}$  vectors ( ${}^{100}C_{50}$ ).

# Computational Burden in RI

Computing the exact randomization distribution is not always possible:

- $n = 6$  and  $n_1 = 3 \rightsquigarrow 20$  assignment vectors ( ${}^6C_3$ ).
- $n = 10$  and  $n_1 = 5 \rightsquigarrow 252$  vectors ( ${}^{10}C_5$ ).
- $n = 100$  and  $n_1 = 50 \rightsquigarrow 1.009 \times 10^{29}$  vectors ( ${}^{100}C_{50}$ ).
- Workaround: sampling!
  - Take  $K$  samples from the treatment assignment.
  - Compute the randomization distribution in the  $K$  samples.
  - Tests are no longer exact, but bias is under your control!  
(we have control over  $K$ )

# Other Test Statistics

- The difference in means is great for when effects are:
  - constant (i.e.,  $\tau = \tau_i$  for all  $i$ ) and additive (e.g.,  $Y(1) - Y(0)$  and not  $Y(1)/Y(0)$ )
  - few outliers in the data
- Outliers  $\rightsquigarrow$  more variation in the randomization distribution.
- What about alternative test statistics?

# Transformations

- What if there was a constant multiplicative effect:  $Y_i(1)/Y_i(0) = C$ ?
- $T_{\text{diff}}$  will have low power in this case as it does not capture the nature of the causal effect.
- $\rightsquigarrow$  Transform the observed outcome using the natural logarithm:

$$T_{\log} = \left| \frac{1}{n_1} \sum_{i=1}^N D_i \log(Y_i) - \frac{1}{n_0} \sum_{i=1}^N (1 - D_i) \log(Y_i) \right| \quad (5)$$

- Log-transforming the outcomes linearizes the multiplicative nature of the causal effect.
- Recall  $\log\left(\frac{a}{b}\right) = \log(a) - \log(b)$ .
- Also useful for skewed distribution of the outcomes.

p.s., See [Chen & Roth \(2024, QJE\)](#): log-like transformations, e.g.,  $\log(1 + Y)$ , with zero-valued outcomes  $\rightsquigarrow$  ATE is arbitrarily sensitive to units of  $Y$ .

# Difference in Median/Quantile

- To protect against outliers: quantiles.
- Difference in medians:

$$T_{\text{med}} = |\text{med}(\mathbf{Y}_t) - \text{med}(\mathbf{Y}_c)| \quad (6)$$

- where  $\mathbf{Y}_t = Y_i$  for  $i : D_i = 1$  and  $\mathbf{Y}_c = Y_i$  for  $i : D_i = 0$
- Recall that the median is the 0.5 quantile.
- Could use other quantiles (e.g., the 0.25 quantile or the 0.75 quantile).

# Two-Sided or One-Sided?

- So far, we have defined all test statistics as **absolute values**.
- $\rightsquigarrow$  testing against a **two-sided** alternative hypothesis:

$$H_0 : \tau_i = 0 \ \forall i \qquad H_1 : \tau_i \neq 0 \text{ for some } i$$

# Two-Sided or One-Sided?

- So far, we have defined all test statistics as **absolute values**.
- $\rightsquigarrow$  testing against a **two-sided** alternative hypothesis:

$$H_0 : \tau_i = 0 \ \forall i \quad H_1 : \tau_i \neq 0 \text{ for some } i$$

- What about a **one-sided** alternative?

$$H_0 : \tau_i = 0 \ \forall i \quad H_1 : \tau_i > 0 \text{ for some } i$$

- For these, use a test statistic that is bigger under the alternative:

$$T_{\text{diff}}^* = \overline{Y}_t - \overline{Y}_c$$

- No absolute value  $\rightsquigarrow$  only large *positive* values count as evidence.



# Other Sharp Nulls

- Sharp null of no effect is not the only sharp null.
- Sharp null in general is one of a **constant additive effect**:  $H_0 : \tau_i = 0.2$ .
  - Implies that  $Y_i(1) = Y_i(0) + 0.2$ .
  - Can still calculate all the potential outcomes!

# Other Sharp Nulls

- Sharp null of no effect is not the only sharp null.
- Sharp null in general is one of a **constant additive effect**:  $H_0 : \tau_i = 0.2$ .
  - Implies that  $Y_i(1) = Y_i(0) + 0.2$ .
  - Can still calculate all the potential outcomes!
- More generally, we could have  $H_0 : \tau_i = \tau_0$  for a fixed  $\tau_0$ .

# Testing Non-Zero Sharp Nulls

- Suppose that we had:  $H_0 : \tau_i = Y_i(1) - Y_i(0) = 1$

| Unit    | Email<br>$D_i$ | Donation<br>$Y_i$ | $Y_i(0)$ | $Y_i(1)$ | Adjusted<br>$Y_i - D_i\tau_0$ |
|---------|----------------|-------------------|----------|----------|-------------------------------|
| Brian   | 1              | 3                 | (2)      | 3        | 2                             |
| Desi    | 1              | 5                 | (4)      | 5        | 4                             |
| Medjine | 1              | 0                 | (-1)     | 0        | -1                            |
| Natasha | 0              | 4                 | 4        | (5)      | 4                             |
| Fifi    | 0              | 0                 | 0        | (1)      | 0                             |
| Matthew | 0              | 1                 | 1        | (2)      | 1                             |

# Testing Non-Zero Sharp Nulls

- Suppose that we had:  $H_0 : \tau_i = Y_i(1) - Y_i(0) = 1$

| Unit    | Email<br>$D_i$ | Donation<br>$Y_i$ | $Y_i(0)$ | $Y_i(1)$ | Adjusted<br>$Y_i - D_i\tau_0$ |
|---------|----------------|-------------------|----------|----------|-------------------------------|
| Brian   | 1              | 3                 | (2)      | 3        | 2                             |
| Desi    | 1              | 5                 | (4)      | 5        | 4                             |
| Medjine | 1              | 0                 | (-1)     | 0        | -1                            |
| Natasha | 0              | 4                 | 4        | (5)      | 4                             |
| Fifi    | 0              | 0                 | 0        | (1)      | 0                             |
| Matthew | 0              | 1                 | 1        | (2)      | 1                             |

- Assignments will now affect  $Y_i$ .
- Solution: use **adjusted outcomes**,  $Y_i^* = Y_i - D_i\tau_0$ .

# Testing Non-Zero Sharp Nulls

- Suppose that we had:  $H_0 : \tau_i = Y_i(1) - Y_i(0) = 1$

| Unit    | Email<br>$D_i$ | Donation<br>$Y_i$ | $Y_i(0)$ | $Y_i(1)$ | Adjusted<br>$Y_i - D_i\tau_0$ |
|---------|----------------|-------------------|----------|----------|-------------------------------|
| Brian   | 1              | 3                 | (2)      | 3        | 2                             |
| Desi    | 1              | 5                 | (4)      | 5        | 4                             |
| Medjine | 1              | 0                 | (-1)     | 0        | -1                            |
| Natasha | 0              | 4                 | 4        | (5)      | 4                             |
| Fifi    | 0              | 0                 | 0        | (1)      | 0                             |
| Matthew | 0              | 1                 | 1        | (2)      | 1                             |

- Assignments will now affect  $Y_i$ .
- Solution: use **adjusted outcomes**,  $Y_i^* = Y_i - D_i\tau_0$ .
- Now, just test sharp null of no effect for  $Y_i^*$ .
  - $Y_i^*(1) = Y_i(1) - 1 \times 1 = Y_i(0)$
  - $Y_i^*(0) = Y_i(0) - 0 \times 1 = Y_i(0)$
  - $\tau_i^* = Y_i^*(1) - Y_i^*(0) = 0$

# Point Estimates via RI

- Is it possible to get point estimates?
- Not really the point of RI, but still possible:
  1. Create a grid of possible sharp null hypotheses.
  2. Calculate p-values for each sharp null.
  3. Pick the value that is “least surprising” under the null.
- Usually this means selecting the  $\tau_0$  with the **highest p-value**.

# Using Regression in RI

- We can also use covariates to improve power.
- For instance, run an OLS regression:

$$(\hat{\beta}_0, \hat{\beta}_D, \hat{\beta}_X) = \arg \min_{\beta_0, \beta_D, \beta_X} \sum_{i=1}^n (Y_i - \beta_0 - \beta_D \cdot D_i - \beta_X \cdot X_i)^2$$

and use  $T_{\text{ols}} = \hat{\beta}_D$  as our test statistic.

# Using Regression in RI

- We can also use covariates to improve power.
- For instance, run an OLS regression:

$$(\hat{\beta}_0, \hat{\beta}_D, \hat{\beta}_X) = \arg \min_{\beta_0, \beta_D, \beta_X} \sum_{i=1}^n (Y_i - \beta_0 - \beta_D \cdot D_i - \beta_X \cdot X_i)^2$$

and use  $T_{\text{ols}} = \hat{\beta}_D$  as our test statistic.

- RI is justified **even if the model is wrong!**
  - OLS is just another way to generate a scalar test statistic.
  - Inference comes from randomization, not from distributional assumptions.



# Using Regression in RI

- We can also use covariates to improve power.
- For instance, run an OLS regression:

$$(\hat{\beta}_0, \hat{\beta}_D, \hat{\beta}_X) = \arg \min_{\beta_0, \beta_D, \beta_X} \sum_{i=1}^n (Y_i - \beta_0 - \beta_D \cdot D_i - \beta_X \cdot X_i)^2$$

and use  $T_{\text{ols}} = \hat{\beta}_D$  as our test statistic.

- RI is justified **even if the model is wrong!**
  - OLS is just another way to generate a scalar test statistic.
  - Inference comes from randomization, not from distributional assumptions.
- If the model is predictive of POs, then  $T_{\text{ols}}$  will have **higher power**.
  - Intuition: controlling for  $X_i$  reduces residual variance  $\rightsquigarrow$  easier to detect smaller effects.

# Next Up

- Inference of the Average Treatment Effect (ATE)
- J. Neyman's way of thinking about the question of causal effect.

## 4/ In-Class Exercise

# In-Class Exercise: Randomization Inference

A small online retailer A/B tested a new product page on 10 customers (5 treated, 5 control;  $Y$  = number of items purchased). **This R script** implements RI with  $T_{\text{diff}} = |\bar{Y}_t - \bar{Y}_c|$  (two-sided p-value = 0.95).

1. **Different test statistic.** Replace diff-in-means with **diff-in-medians**:  
 $T_{\text{med}} = |\text{med}(Y_t) - \text{med}(Y_c)|$ . Compare the p-value to the baseline.  
Look at the data, why do they differ?
2. **One-sided test.** Using your diff-in-medians code from Task 1, remove the absolute value from **both**  $T^{\text{obs}}$  and the randomization distribution.  
Compare the p-value to Task 1. Why did it change?
3. **Changing an outcome.** Pick one person and change their  $Y$  value.  
Re-run RI. Which of the following changed, and why?
  - (i) The set of possible treatment assignments
  - (ii) The randomization distribution of  $T_{\text{diff}}$
  - (iii) The observed test statistic  $T^{\text{obs}}$

**Have a great weekend! :)**

*Contact Information:*

[jaewon.yoo@iss.nthu.edu.tw](mailto:jaewon.yoo@iss.nthu.edu.tw)

<https://j1yoo.github.io/>

